

READING TOOL

Make a chart that lists and describes the different steps of scientific methodology.



INTERACTIVITY

Try your hand at a simulation of a scientific investigation.

Observing and Asking Questions Scientific investigations begin with **observation**, the act of noticing and describing events or processes in a careful, orderly way. Of course, scientific observation involves more than just “looking.” The right kind of observation leads to asking questions that no one has asked (or answered) before.

Inferring and Forming a Hypothesis After posing questions, scientists use further observations to make inferences. An **inference** is a logical interpretation based on what scientists already know. Inference, combined with a creative imagination, can lead to a hypothesis. A **hypothesis** is a tentative scientific explanation that can be tested by further observation or by experimentation. Scientific hypotheses must then be tested by gathering data that can either support or reject them.

Designing Controlled Experiments Testing hypotheses often involves designing experiments that measure factors that can change. These changeable factors are called variables. Some possible variables include temperature, light, time, and availability of nutrients. Ideally, a hypothesis should be tested by an experiment in which only one variable is changed. All other variables should be kept unchanged, or controlled. This is called a **controlled experiment**.

Controlling Variables It is important to control variables because if several variables are changed, researchers can’t easily tell which variable is responsible for any results they observe. The variable deliberately changed is called the **independent variable**. The variable that is observed and that changes in response to the independent variable is called the **dependent variable**.

Control and Experimental Groups Typically, experiments are divided into control and experimental groups. A **control group** is exposed to the same conditions as the experimental group except for changes in the independent variable. Scientists always try to reproduce or replicate their observations by setting up several control and experimental groups, rather than just a single pair.

Collecting and Analyzing Data Scientists observe their experiments, gathering two main types of **data**. *Quantitative data* are numbers obtained by counting or measuring. In the marsh grass experiment, quantitative data could include the number of plants per plot; the length, width, and weight of each blade of grass; and so on. *Qualitative data* are descriptive and involve characteristics that cannot usually be measured. Qualitative data in the marsh grass experiment might include notes about whether the grass was growing upright or sideways, or whether one or another plot was disturbed by clam diggers.

Figure 1-3

Collecting Data

This scientist is collecting data on a coral reef in Indonesia.

